Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/14/2010
PI Name:	Prufert-Bebout, Leslie Eileen		
Project Title:	Microbial Assay Technologies for Space (MATS):A Coordinated Ecosystem Response Assay Technology		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	06/01/2001	**End Date:**	05/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

To develop a core capability to produce and manipulate small scale microbial communities as defined “test systems,” and to implement micro-scale technologies (microsensor, molecular) for tracking population, process and product changes in these test systems as a function of experimental manipulations.

A primary objective of this project is the development of an incubation chamber prototype, capable of remote deployment, which will allow for the maintenance and analyses of mixed population photosynthetic biofilms and microbial mats. This chamber will be designed and built to allow for the use of various imaging, sampling, microsensor and molecular techniques to assess the microbial community population responses to various experimental manipulations. A second objective is to modify natural microbial communities, specifically selecting for attributes which will allow the community to be incorporated as a complex biological test-system for specific use with this hardware.

Task Description:	Among NASA’s primary goals are the imperatives to further our knowledge of how life from Earth can adapt to space conditions. Life on Earth is dominated by microbes, in terms of biomass, overall rates of activity, use of potentially available habitats, and length of time present on the planet. Microbes are intimately associated with humans and are necessary for regenerative cycling of energy and elements in enclosed systems (space) just as they are on the Earth. It is essential to NASA’s space exploration goals to achieve an advanced capability to understand, manage and manipulate microbial ecosystems, particularly as we attempt to explore beyond the Earth’s biosphere. A crucial element of this exploration is to recognize that microorganisms rarely exist in isolation, but rather live, interact and evolve in highly interdependent, complex communities such as microbial mats. Physiological properties of microorganisms are frequently very different in intact communities than in isolated cultures, and furthermore the linkages of nutrients and chemical energy flow between the various members of the community cannot be studied within single community components in isolation. Therefore in order to realize the highest scientific return germane to NASA’s goals, it is imperative to study microorganisms, their interactions and co-dependencies with surrounding populations and with gross environmental conditions, at the community level. The goal of this project is to develop both a complex biological system and an innovative incubation system for that microbial community which provides for whole ecosystem monitoring, in a highly dense, compact system, so that rapid changes in community composition or function can be detected.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received this reporting period. [Ed. note: FY2004 record added in December 2010 for statistical reporting purposes]
Bibliography Type:	Description: (Last Updated: 02/27/2004)